2023 AMC 10B Problem 17 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 17. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 17

Problem:

A rectangular box $\mathcal{P}$ has distinct edge lengths a, b, and c. The sum of the lengths of all 12 edges of $\mathcal{P}$ is 13 , the sum of the areas of all 6 faces of $\mathcal{P}$ is $\dfrac{11}{2}$, and the volume of $\mathcal{P}$ is $\dfrac{1}{2}$. What is the length of the longest interior diagonal connecting two vertices of $\mathcal{P}$?

Answer Choices:

A. 2

B. $\dfrac{3}{8}$

C. $\dfrac{9}{8}$

D. $\dfrac{9}{4}$

E. $\dfrac{3}{2}$

Solution:

The longest diagonal passes through the interior of the box, connecting opposite vertices; its length is $\sqrt{a^2+b^2+c^2}$. Because $4a+4b+4c=13$ and $2ab+2bc+2ac=\frac{11}{2}$, it follows that

$$\left(\frac{13}{4}\right)^2 = (a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ac = a^2+b^2+c^2+\frac{11}{2}$$

Therefore

$$a^2+b^2+c^2 = \left(\frac{13}{4}\right)^2 - \frac{11}{2} = \frac{81}{16}$$

The longest diagonal has length $\sqrt{a^2+b^2+c^2} = \text{(D)}\boxed{\frac{9}{4}}$.

Note: Such a box $\mathcal{P}$ exists with edge lengths $\frac{1}{4}, 1$, and 2 . The volume is not needed to solve this problem.